University of Antwerp
Faculty of Applied Engineering

Lab 5 – Replication

Distributed Systems

Andreas Gavrielides

University of Antwerp – imec, IDLab – Faculty of Applied Engineering, Belgium

imec
embracing a better life

Overview of lab sessions

- **Lab 1: DONE**
  - Introductory lab & TCP/UDP
  - Setup Remote nodes
- **Lab 2: DONE**
  - REST protocol
  - Remote nodes
  - 2 sessions
- **Lab 3: Project: DONE**
  - Naming server
  - Remote nodes
  - 2 sessions
- **Lab 4: Project -> In progress**
  - Discovery service
  - Remote nodes
  - 2 sessions
- **Lab 5: Project -> Today**
  - Replication
  - Remote nodes
  - 4 sessions
- **Lab 6: Project**
  - Agents
  - Remote nodes
  - 1 session
- **Lab 7: Project**
  - GUI
  - Remote nodes
  - 1 session
- **Finishing project**
  - Remote nodes
  - 4 sessions

Deadlines for Lab Reports

Lab report	Deadline
Lab 1 – Introduction and TCP/UDP	15/03/2026
Lab 2 – REST protocol	19/03/2026
Lab 3 – Naming Server	19/04/2026
Lab 4 – Discovery	06/05/2026
Lab 5 – Replication	13/05/2026
Lab 6 – Agents	20/05/2026
Lab 7 – GUI	27/05/2026
Final Report	15/06/2026
Peer evaluation	19/06/2026
Final Exam	22/06/2026

Lab Replication [project]

- **Goal of the session:**
  - To enable `replication of files` in system Y
- **Team work:**
  - Work in your groups (available on Blackboard), **divide the work!**
- **Phases:**
  1. Starting
  2. Update
  3. Shutdown

Phases

- **Starting:**
  - All files that are stored on each node should be replicated to corresponding nodes in system Y. This way, a new node to which the file is replicated becomes the owner of the file.
- **Update:**
  - If new files are added locally to certain node, or deleted from a node, this state has to be synchronized in the whole system. If a new file is added, then it has to be replicated. Otherwise, if deleted, it has to be deleted from the replicated files of the file owner as well.
- **Shutdown:**
  - If needed, all files locally stored on a node that is about to shutdown should be transferred to other nodes. Otherwise, they can be deleted and this change needs to be synced as described above in the Update.

Notation

- **Local files:**
  - Are files that are locally stored on a node
- **Replicated files:**
  - Are files that are replicated to a new node
- **Owner of the files:**
  - Is the node that can be referred to by receiving pointer from Naming server
    - This node keeps track of all locations where the file is stored
- **Replicated node:**
  - Is the node to which the file is replicated

Prepare a fully working file transfer class that can be used to transfer files from one node to another (File transfer using TCP socket)

Phase: Starting

1. **After bootstrap and discovery, the new node has to verify its local files (folder on the hard drive)**

2. **For all files, hash values are calculated and the local node has to report that to the naming server**

3. **Naming server receives hash values of files that need to be replicated. Replication is performed as follows:**
    1. node.hash < file.hash
    2. Node is a replicated node
    3. It becomes the owner of this file and creates a log with information on the file (references for the file)

4. **Naming server informs the originating node on that and then the originating node transfers the files to the replicated nodes.**

Phase: Update

1. When a local file is added, it should be replicated immediately.

2. In order to detect new files being added to the node, a thread can be executed that checks for changes on the local folder at regular a intervals.

Phase: Shutdown (1/3)

1. **When a node is terminated, the files replicated on this node, should be moved to the <u>previous node</u>. This previous node becomes the new <u>owner of the files</u>.**
   - Edge Case:
     - When a file that is replicated to this node is locally stored on the previous node, then the file shouldn’t be transferred to this previous node (it would then store both the local file and the replication of the file), but is should be transferred to the previous node of the previous node.
       (Check example3: if node1 with ID = 5 fails, image5.png with hash = 7 won’t be replicated to node 4 with ID = 28 because ”image5.png” is already its local file. Instead, ”image5.png” will be replicated to node3 with ID = 17)

Phase: Shutdown (2/3)

1. The replicated files must be transferred over TCP

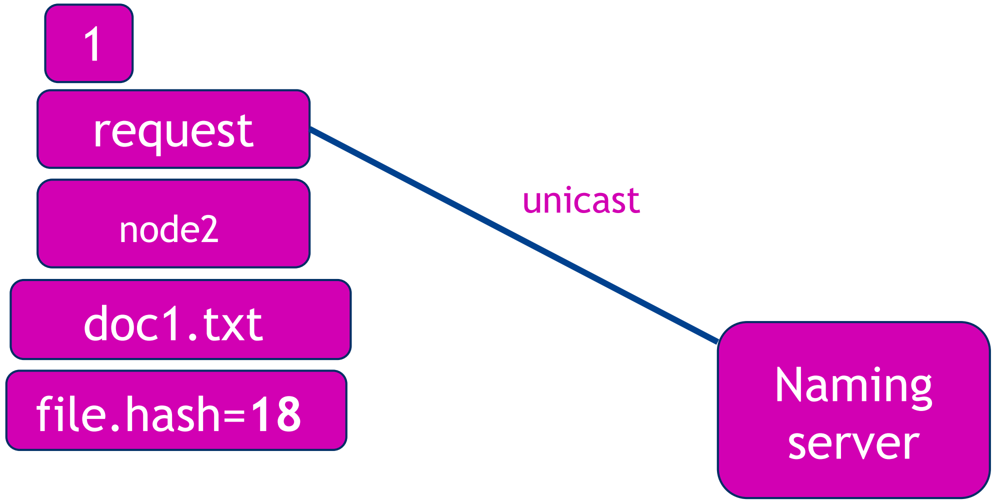
2. The file log of every replicated file, will be transferred to the new node, and updated accordingly.

Phase: Shutdown (3/3)

1. **In case of termination, the owner of the file needs to be warned that the node that stores the local file will be terminated**

2. **Owner of the file:**
    1. In case that the file was never downloaded by other nodes -> file can be removed
    2. Otherwise *download locations* needs to be updated in the file log

Replicaton example 1



public

Replicaton example 1

1

request

node2

doc1.txt

file.hash=18

unicast

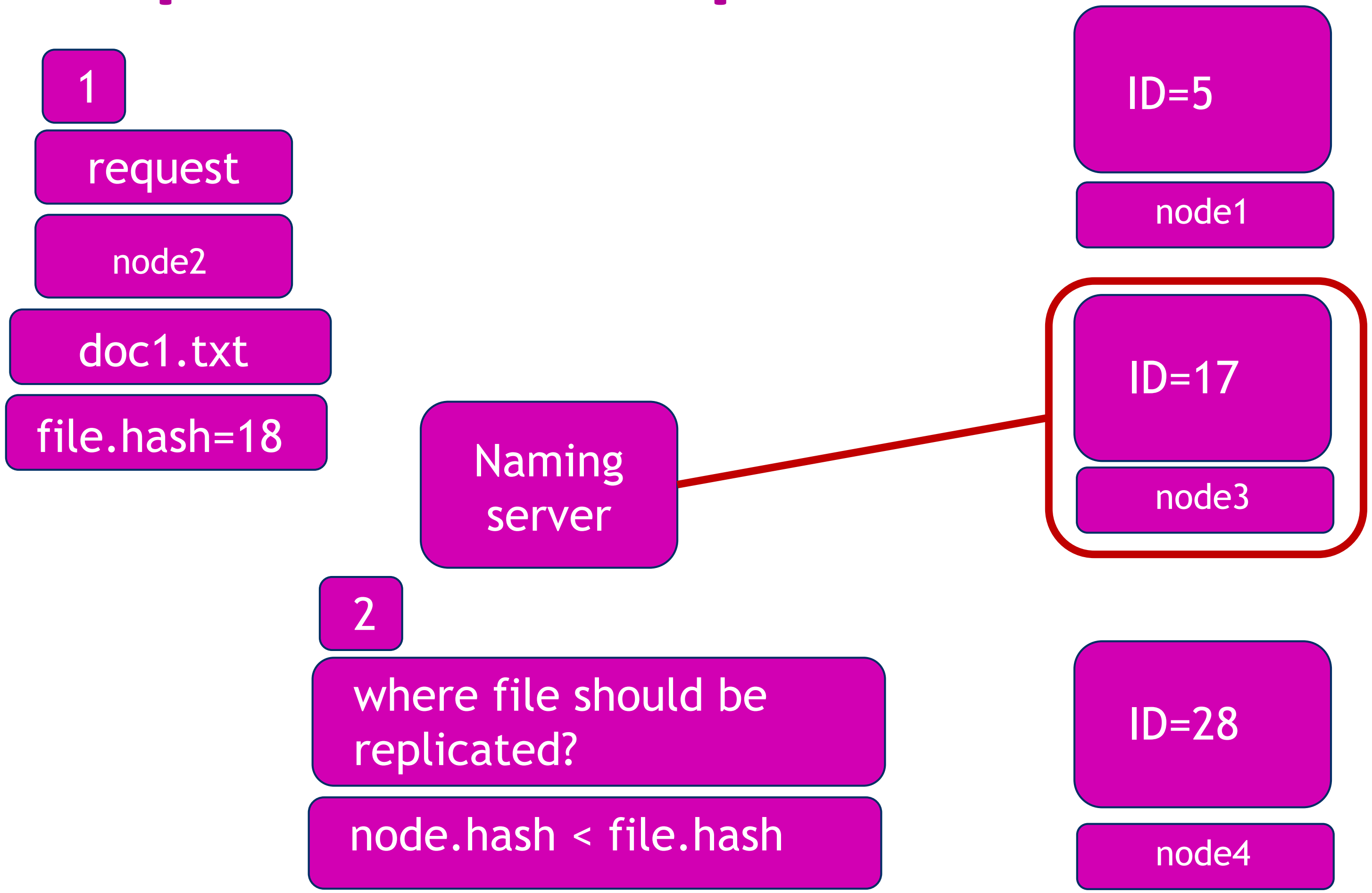
Naming server

2

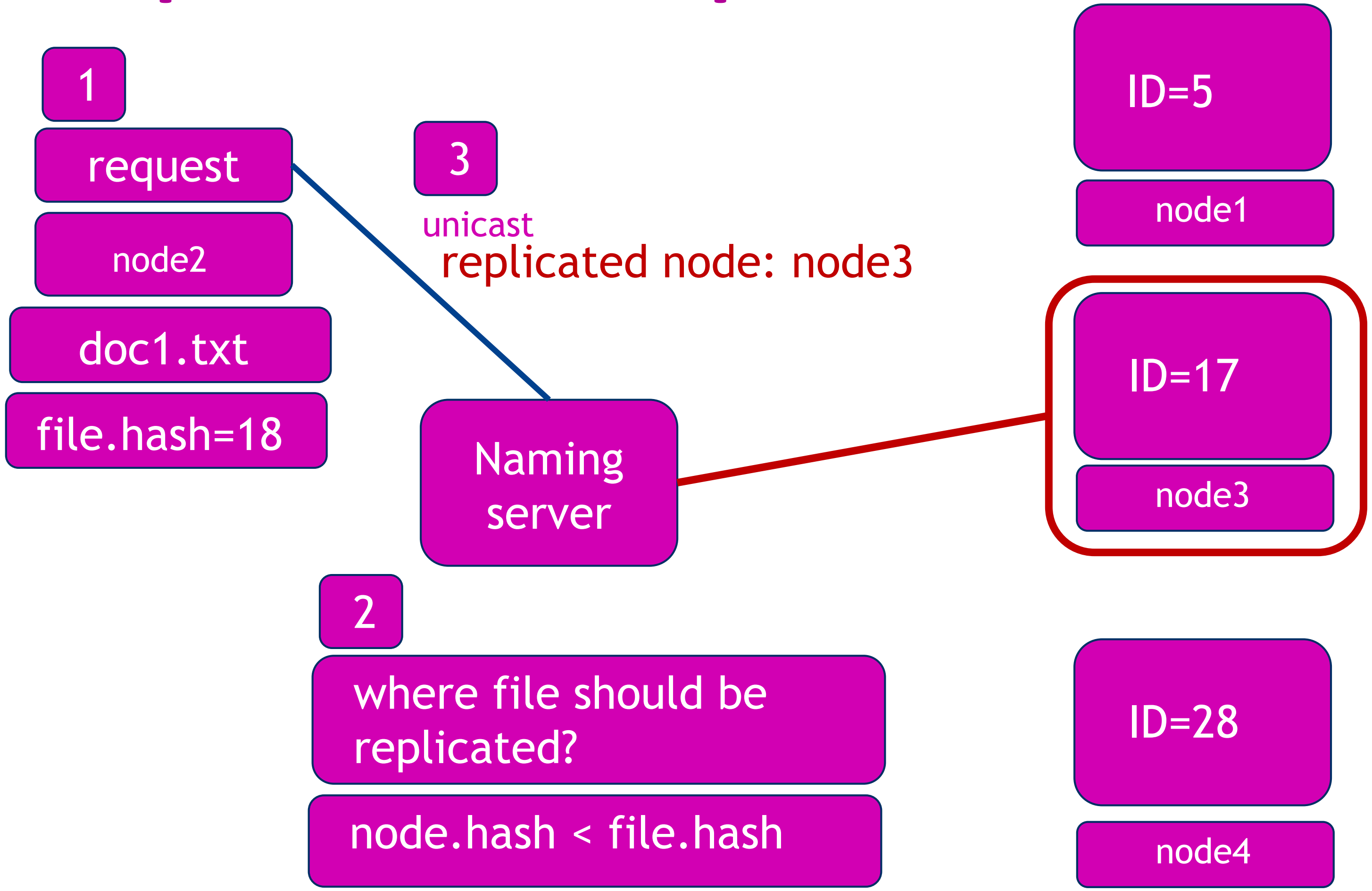
where file should be replicated?

node.hash < file.hash

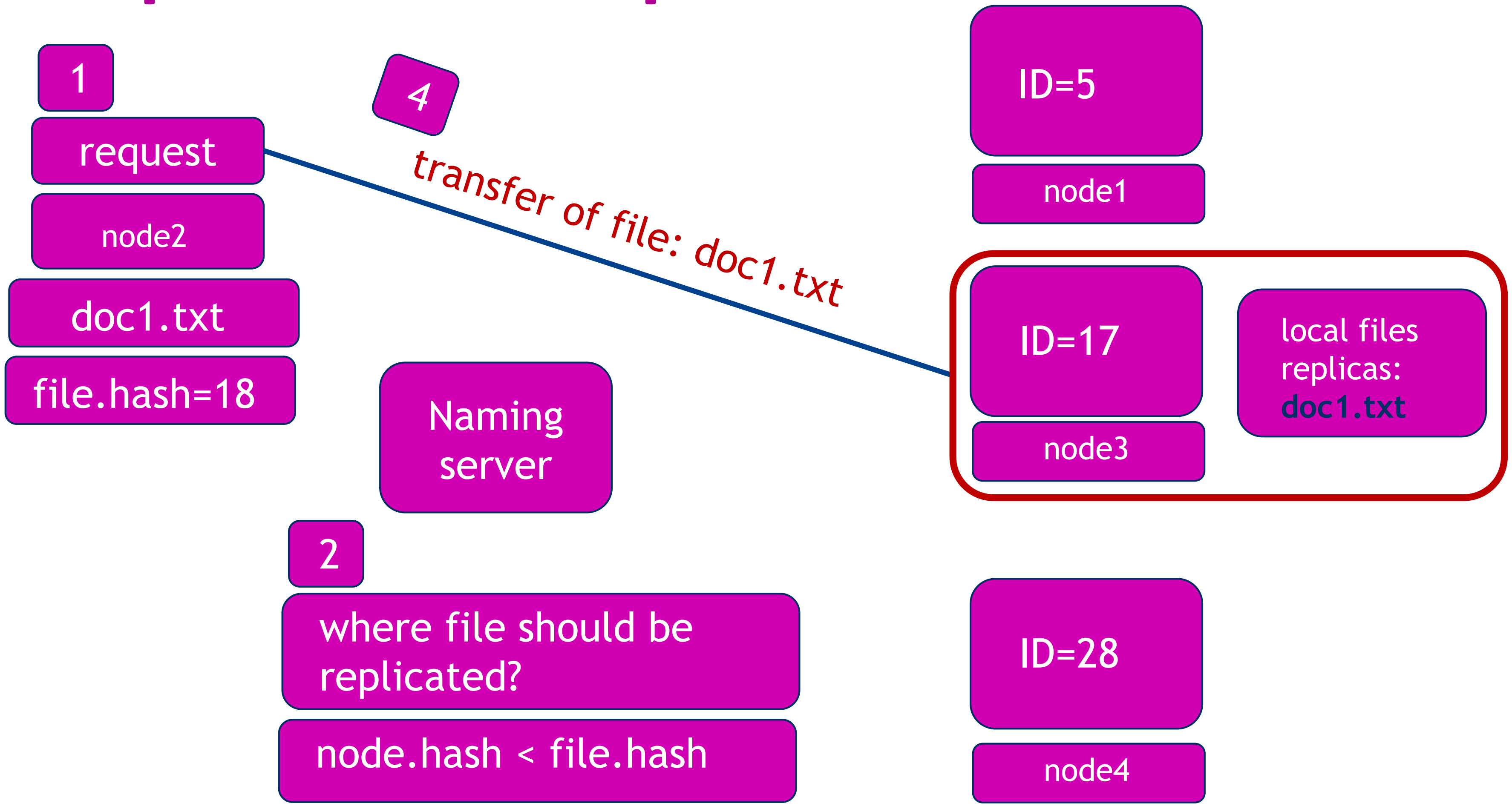
Replicaton example 1



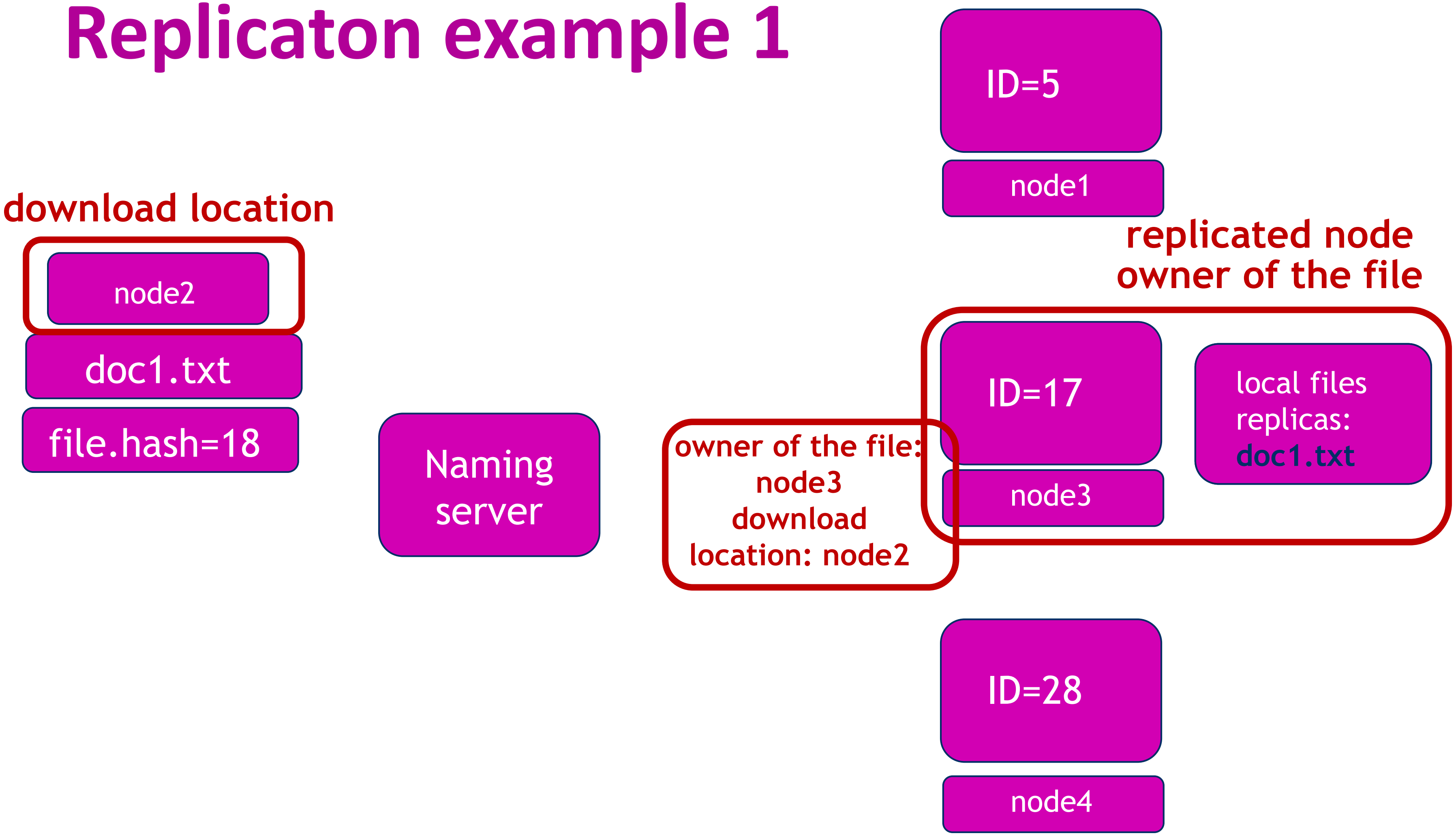
Replicaton example 1



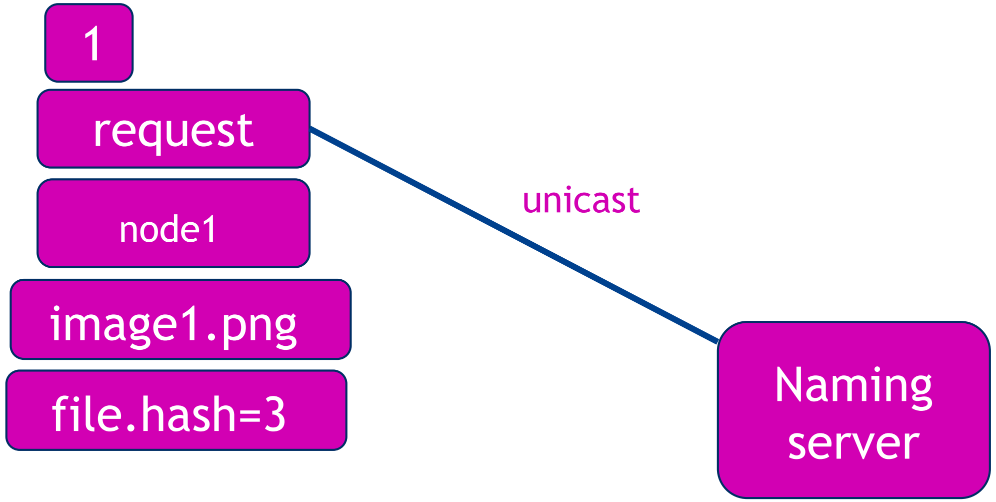
Replicaton example 1



Replicaton example 1



Replicaton example 2



Replicaton example 2

1

request

node1

image1.png

file.hash=3

unicast

Naming server

2

where file should be replicated?

node.hash < file.hash

Replicaton example 2

1

request

node1

image2.png

file.hash=3

3

replicated node: node4

ID=5

node1

ID=17

node3

Naming server

2

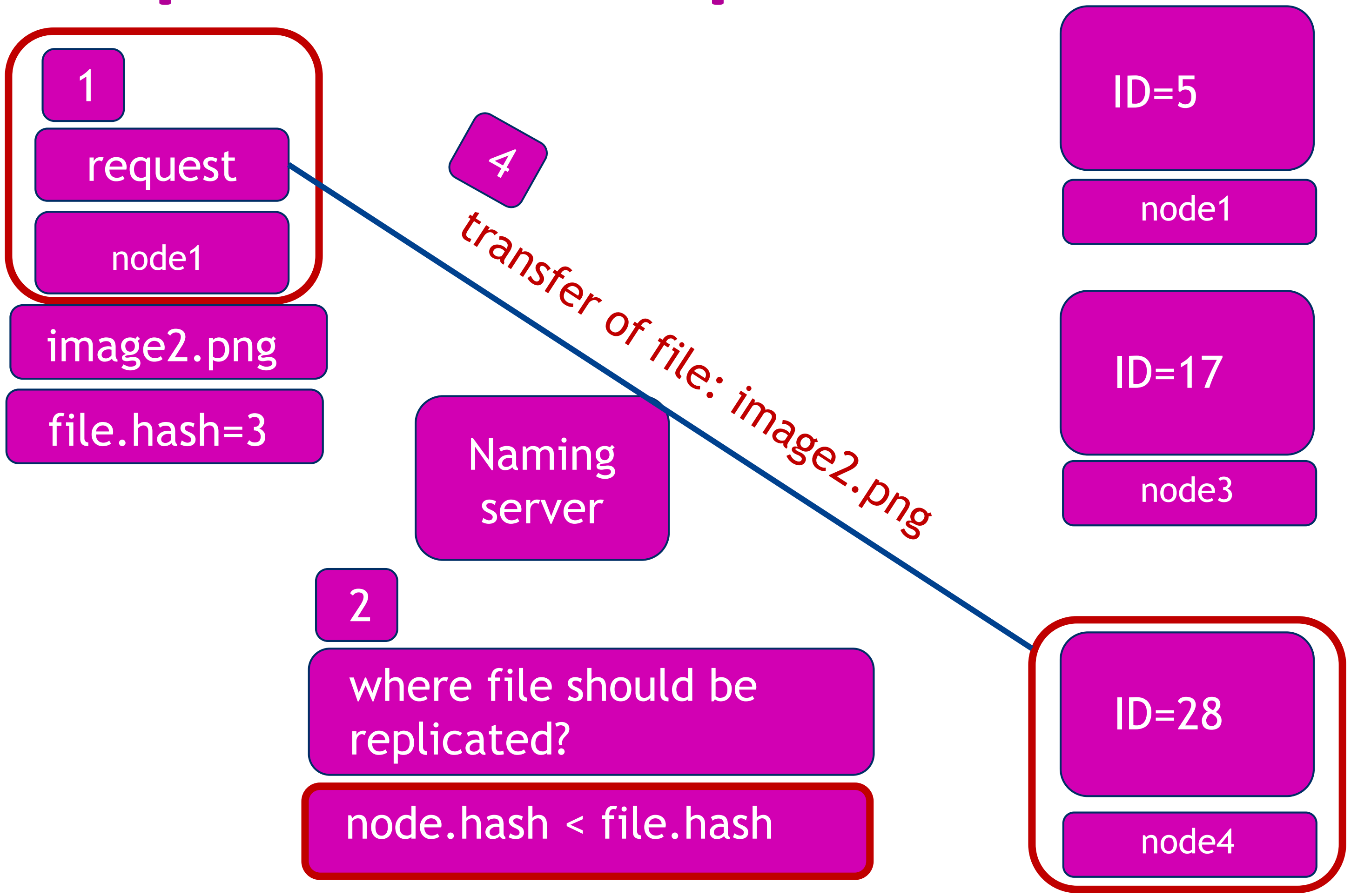
where file should be replicated?

node.hash < file.hash

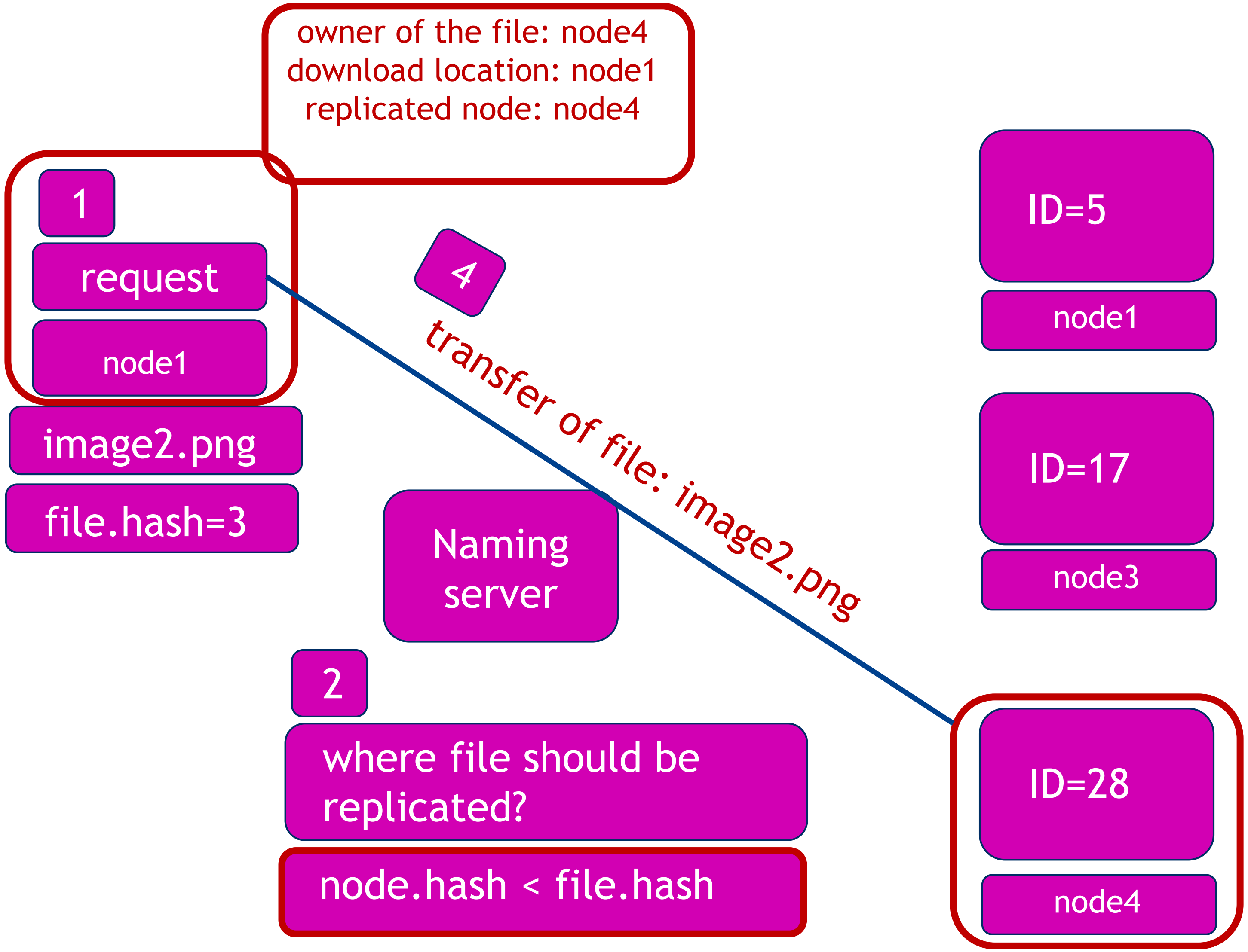
ID=28

node4

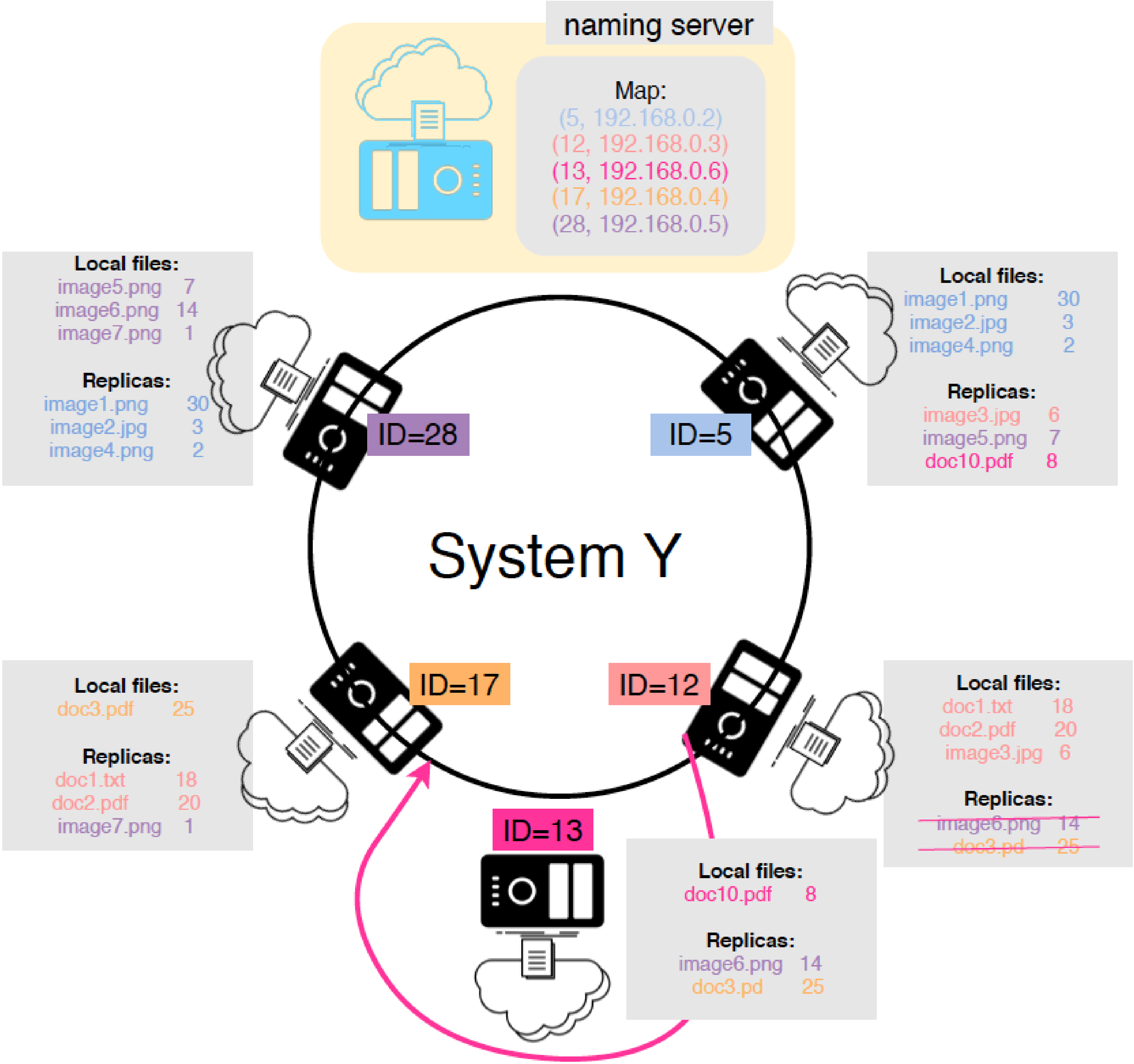
Replicaton example 2



Replicaton example 2



Replicaton example 3



public

Tests

1. Add a node with a number of local files, and verify if the files are replicated to the correct destinations.

2. Add new files to the node. Make sure the updates are synced in the whole system

3. Add a file, where the hash of the file, points to the node that stores the file locally.

4. Remove a node and verify if all replicated files are transferred on the correct nodes. Verify if local files are removed from the network.

Deadlines

- Upload your individual report in pdf to *Blackboard Report Submission tab.*
- on May 15/05/2025 23u59